

Business Model Canvas

Document 2 of 4 — AurumXchange Business Case

Nine-block business model canvas translating the VPC fit into customer segments, value propositions, channels, relationships, revenue streams, key activities, resources, partnerships, and cost structure.

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For Ghana Chamber of Gold Buyers

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The Business Model Canvas translates the VPC's Fit Statement into a sustainable commercial architecture. It is presented here as a single-page narrative across all nine building blocks, organized so that the right side (Customer Segments, Value Propositions, Channels, Customer Relationships, Revenue Streams) describes how AurumXchange creates and captures value in the market, and the left side (Key Activities, Key Resources, Key Partnerships, Cost Structure) describes how Smart Innovations Ghana Limited builds and operates the platform. The Revenue Streams block carries pricing assumptions that reappear, line-for-line, in the Financial Documentation's bottom-up revenue model; the Key Partnerships block lists external dependencies — including the Ghana Gold Board (GoldBod, established April 2025 under Act 1140) — that reappear, line-for-line, in the OpEx budget. This is the Golden Thread in its first explicit form.

1. Customer Segments

AurumXchange serves two primary segments and one secondary segment. The primary segments are the two sides of every trade — licensed sellers and licensed buyers — and the secondary segment is the Chamber itself, which is both regulator-of-record for trade approvals and the recipient of 30% of platform revenue. Each primary segment has a distinct persona; the same person is never both buyer and seller on the same trade, but a single firm may register as both over time. Personas are drawn against the real Ghanaian gold sector: large-scale mining companies (Newmont Ghana, Gold Fields Ghana, AngloGold Ashanti Obuasi, Kinross Chirano, Asanko Gold) operate outside the GoldBod channel under bilateral offtake, while small-scale miners and aggregators now operate within the GoldBod sole-buyer framework established by Act 1140 [UNCTAD, 2025; goldbod.gov.gh].

Table 2.1. Customer Segments & Personas

Segment	Persona	Description	Economic Buyer
Primary — Tier 2 Sellers (small-scale channel)	Abena Owusu (Licensed Small-Scale Miner / Aggregator)	39-year-old operator of a registered small-scale mining cooperative in Tarkwa; holds Minerals Commission licence; since April 2025 sells 3–8 kg/month of dore through GoldBod as the sole legal buyer of small-scale-mined gold [GoldBod Act 1140].	Abena's cooperative (SME)

Primary — Tier 2 Buyers	Kofi Mensah (Licensed Exporter-Aggregator)	41-year-old MD of a Kumasi-based exporter; holds Minerals Commission licence + PMMC LGE permit; sources 4 kg/month from large-scale mining companies under bilateral offtake (outside GoldBod's mandate); exports to refiners in Switzerland and UAE.	Kofi's firm (SME)
Primary — Tier 1 Buyers	Marcus Weber (Refiner Procurement Lead)	47-year-old Head of Procurement at a Swiss LBMA-accredited refiner; sources 50–200 kg/month of West African dore; needs responsible-origin documentation satisfying the LBMA Responsible Gold Guidance five-step due diligence [lbma.org.uk, 2024].	Refiner corporate parent (Enterprise)
Secondary — The Chamber	Nana Chairperson (Chamber President)	Authorized representative of the Ghana Chamber of Gold Buyers; sets trading rules, approves member admissions, approves each trade above a threshold; receives 30% revenue share.	The Chamber (Association)

Table 2.1. Four personas reflecting the post-April-2025 dual-channel environment: small-scale gold via GoldBod (Abena); large-scale gold via bilateral offtake (Kofi). Tier 1 buyers (Marcus) require LBMA RGG compliance.

2. Value Propositions

The Value Propositions block summarizes the VPC Fit Statement, restated once per segment where the proposition differs. All four propositions are grounded in the same compliance-gated exchange architecture; the differences are in the specific pain or gain each segment prioritizes.

Table 2.2. Value Propositions by Segment

Segment	Value Proposition
Tier 2 Sellers — Abena (small-scale, post-GoldBod)	AurumXchange gives you verified, licensed buyers in one place and ensures your gold is paid for via escrow before title transfers — while also routing your small-scale-mined gold through the GoldBod sole-buyer workflow in compliance with Act 1140. Your funds-at-risk window collapses and your realized price is exposed to competitive bids rather than a single GoldBod reference price.
Tier 2 Buyers — Kofi (exporter-aggregator)	AurumXchange gives you a single compliance-gated workflow where every seller is KYC-verified, every lot has a tamper-evident audit trail aligned to the LBMA Responsible Gold Guidance, and every payment is held in escrow (with Stanbic Bank Ghana or Ecobank Ghana) until title transfers — replacing 6 days of export-documentation drag across PMMC, GRA Customs, and Bank of Ghana forex verification with one auditable trade.
Tier 1 Buyers — Marcus (LBMA refiner)	AurumXchange gives you LBMA RGG-aligned responsible-origin documentation and a complete chain-of-custody audit trail for every kilogram of Ghanaian dore you source — satisfying your refiner's OECD Due Diligence obligations without adding procurement overhead.
The Chamber	AurumXchange gives you a single digital surface to manage member licensing, approve trades, and earn 30% of platform revenue — without investing upfront capital or operating the technology yourself. The platform absorbs the GoldBod workflow introduced by Act 1140 (April 2025), reducing member friction during the regulatory transition.

Table 2.2. Four propositions, all restatements of the VPC Fit Statement tuned to each persona's primary pain or gain. The GoldBod workflow is explicitly absorbed into the seller and Chamber propositions.

3. Channels

Channels describe how a user discovers, evaluates, purchases, and accesses AurumXchange. Because the Chamber is the channel partner, the discovery and evaluation channels are largely owned by the Chamber rather than by Smart Innovations — this is what makes the 30% revenue share defensible. The access channel is owned by Smart Innovations and is the only channel that incurs variable per-user cost.

- **Discovery:** Chamber membership communications (quarterly meeting, member WhatsApp groups, Chamber website), Chamber-led industry events including the Ghana Mining & Energy Conference (GMEC) and the West African Mining & Power Conference, and word-of-mouth between licensed buyers. Smart Innovations does not run paid acquisition in the discovery channel — the Chamber's endorsement is the acquisition.
- **Evaluation:** Chamber-hosted demo sessions (in-person in Accra, Kumasi, Tarkwa, Obuasi), on-demand product walkthrough video, and a 14-day free Professional-tier trial for verified members.
- **Purchase / Subscription:** Self-service upgrade inside the platform (Free → Professional → Enterprise), with invoice billing for Enterprise. No salesperson touches the transaction below the Enterprise tier. Payment via mobile money (MTN MoMo, Telecel Cash) or bank transfer (Ghana Interbank Payment Settlement System, GHIPSS).
- **Access:** Web application (responsive, mobile-friendly), accessible from any modern browser. Native mobile apps (iOS/Android) are explicitly Out-of-Scope for MVP and are deferred to a post-launch Phase 5 release.

4. Customer Relationships

Different segments expect different relationships, and the platform must support all four simultaneously without forcing the higher-touch relationships onto users who do not want them. Self-service is the default; dedicated account management is reserved for Enterprise subscribers and the Chamber itself.

- **Tier 2 Sellers & Buyers:** Self-service with automated onboarding (KYC via Smile Identity, Minerals Commission licence verification, GoldBod registration check for small-scale gold traders). In-app chat support with a 4-hour first-response SLA during business hours (08:00–18:00 GMT).

- **Tier 1 Buyers (Enterprise):** Dedicated account manager from Smart Innovations Ghana Limited, quarterly business review, dedicated Slack channel, and direct line to the product team for feature requests.
- **The Chamber:** Strategic partnership relationship — monthly steering committee with Smart Innovations' CEO and Head of Product, joint roadmap planning, and shared quarterly reporting on trade volume, revenue, and compliance incidents.
- **Community:** All members are invited to a Chamber-moderated community forum (separate from trading) for industry discussion; this is owned by the Chamber, not by Smart Innovations.

5. Revenue Streams

Revenue flows from platform users, not from the Chamber. The structure is deliberately hybrid: a small per-side transaction fee that scales with trade value without discouraging large trades, recurring subscription revenue that is independent of trading volume, and value-added service fees layered on top for users who opt in. The exact numbers below reappear unchanged in the Financial Documentation's bottom-up model — this is the single most important Golden Thread linkage.

Table 2.3. Revenue Streams — Pricing Assumptions (Golden Thread Anchor)

Revenue Source	Price	Charged To	Frequency
Buyer transaction fee	0.10% of trade value	Buyer	Per trade
Seller transaction fee	0.10% of trade value	Seller	Per trade
Platform service fee	Included in above	Both parties	Per trade
Premium membership — Professional	US\$100/month	Businesses	Monthly
Premium membership — Enterprise	US\$500+/month (tiered)	Businesses	Monthly
Featured supplier listing	US\$250/month	Sellers (opt-in)	Monthly
RFQ submission fee (optional)	US\$5/RFQ	Buyers (opt-in)	Per RFQ
Digital contract fee	US\$25/contract	Transaction parties	Per trade
Escrow fee	0.05% of trade value	Transaction parties	Per trade

Logistics coordination fee (Phase 3)	US\$150/shipment	Opt-in users	Per shipment
Trade finance / insurance referral commission	5–10% of premium	Financial institutions / insurers	Per referral

Table 2.3. Every line in this table reappears in the Financial Documentation's bottom-up revenue model (Table 3.5). Changing a price here requires changing it there.

Worked example grounded in late-2024 LBMA prices: a buyer purchases 10 kg of gold valued at approximately US\$1,000,000 (10 kg × ~US\$100,000/kg dore equivalent at LBMA PM Fix in the US\$2,500–2,800/oz range during late 2024 [LBMA, 2024]). At a 0.10% fee on each side, the platform earns US\$1,000 from the buyer and US\$1,000 from the seller — US\$2,000 in transaction fees. The escrow fee adds $0.05\% \times \text{US\$1,000,000} = \text{US\$500}$ (split between the parties), and the digital contract fee adds US\$25. The platform's total revenue on this single trade is therefore US\$2,525. Under the 70/30 revenue split, US\$1,767.50 goes to Smart Innovations Ghana Limited and US\$757.50 goes to the Chamber. This is the unit economics the Financial Documentation projects across thousands of trades per year.

6. Key Activities

Key Activities are what Smart Innovations Ghana Limited must do daily and weekly to build and run AurumXchange. They are deliberately categorized so that the Cost Structure block can later assign a cost centre to each.

- **Agile development:** Two-week sprint cycle; backlog grooming, sprint planning, daily standups, sprint review and retrospective. Cross-functional team of frontend, backend, mobile-web, and platform engineers based in Accra with satellite presence in Kumasi for member proximity.
- **DevOps & SRE:** CI/CD pipeline maintenance, infrastructure-as-code, observability (Datadog, Sentry), on-call rotation, incident response with a 30-minute P1 acknowledgement SLA. AWS Ghana region (af-west-1) preferred for data residency per Ghana Data Protection Act 2012 (Act 843).
- **Quality assurance:** Manual exploratory testing plus automated regression suite; security testing via Snyk and quarterly third-party penetration test; SOC 2 control evidence collection.
- **Member support & success:** Tier 1 support via in-app chat (4-hour SLA), Tier 2 escalation to product engineering, proactive success outreach for Professional and Enterprise subscribers.
- **Compliance operations:** Ongoing KYC/AML monitoring via Smile Identity (operating in 50+ African countries [Smile Identity, 2024]), sanctions list updates, suspicious-activity report

handling with the Financial Intelligence Centre (FIC, established under AML Act 2020 Act 1044 [OGP commitment GH0029]), quarterly compliance review with the Chamber and GoldBod.

- **Data processing:** Nightly ETL for analytics dashboards, weekly audit-log archival to write-once storage, monthly data-quality reconciliation against PMMC and GoldBod records.

7. Key Resources

Key Resources are what Smart Innovations Ghana Limited must own or control to make the model work. The most important are the team and the audit-log IP — not, notably, the gold itself, which the platform never touches.

- **Engineering team:** 12 FTE at MVP launch (1 CTO, 1 Head of Product, 5 engineers, 2 QA, 1 designer, 1 DevOps, 1 PM); grows to 16 FTE by Year 3. Locally hired in Ghana with selective senior expatriate contracts for the CTO and Lead Backend Engineer during Year 1.
- **Cloud infrastructure:** AWS Ghana region (af-west-1, generally available since 2021) preferred for data residency under Ghana Data Protection Act 2012 (Act 843); disaster-recovery failover to AWS South Africa (af-south-1, Cape Town).
- **Proprietary audit-log architecture:** Append-only, hash-chained trade state log pre-validated against the LBMA Responsible Gold Guidance five-step due diligence [lbma.org.uk, 2024]; the design IP that makes the platform's compliance claim defensible and is the single largest source of competitive moat.
- **Chamber partnership contract:** The 5-year exclusive revenue-sharing agreement with the Ghana Chamber of Gold Buyers, with automatic renewal subject to performance KPIs.
- **Brand trust:** The Chamber's endorsement is the single largest brand asset; Smart Innovations Ghana Limited's brand is secondary and grows through delivered reliability.

8. Key Partnerships

Key Partnerships are external dependencies without which the platform cannot operate. Every partner in this block appears as a line item in the OpEx budget — this is the second explicit Golden Thread linkage. The list below reflects the post-April-2025 reality: GoldBod is now a critical partner for any trade touching small-scale-mined gold.

Table 2.4. Key Partnerships & OpEx Mapping (Golden Thread Anchor)

Partner	Role	OpEx Line Item
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Ghana Gold Board (GoldBod, Act 1140)	Sole legal buyer of small-scale-mined gold since 2 April 2025 [UNCTAD, 2025]; integration mandatory for small-scale channel trades	Compliance integration fee (Table 3.2, row 6)
AWS Ghana (af-west-1)	Cloud hosting with data residency per DPA 2012 (Act 843)	Cloud hosting & database (Table 3.2, row 1)
Smile Identity	KYC/AML verification at onboarding and ongoing; operates in 50+ African countries [Smile Identity, 2024]	Third-party API / SaaS (Table 3.2, row 2)
Stanbic Bank Ghana or Ecobank Ghana	LBMA-aligned escrow account holding and two-signature release [stanbicbank.com.gh; ecobank.com]	Banking integration fee (Table 3.2, row 3)
LBMA price feed (FastMarkets or Refinitiv)	LBMA PM Fix for daily price benchmarking	Third-party API / SaaS (Table 3.2, row 2)
PMMC (Precious Minerals Marketing Company)	Export release workflow (LGE export certification) [pmmc.gov.gh]	Compliance integration fee (Table 3.2, row 6)
Minerals Commission of Ghana	Licence verification API [mincom.gov.gh]	Compliance integration fee (Table 3.2, row 6)
Bank of Ghana	DGPP reference prices; forex verification for export proceeds [Ghana Chamber of Mines PWYP, 2024]	Compliance integration fee (Table 3.2, row 6)
SOC 2 / ISO 27001 auditor	Annual security attestation	Compliance & security maintenance (Table 3.2, row 7)
External legal counsel (Ghana-barred)	Privacy policy, member terms, regulatory liaison with GoldBod and Minerals Commission	Ongoing software licenses & legal (Table 3.2, row 4)
Outsourced QA partner	Quarterly penetration testing	Compliance & security maintenance (Table 3.2, row 7)

Table 2.4. *Eleven Key Partnerships, each mapping to an OpEx line item. GoldBod is now the #1 partner on the list, reflecting the April 2025 regulatory transition; no partnership is listed here without a corresponding budget allocation in Table 3.2.*

9. Cost Structure

Costs are categorized as fixed (do not scale with trade volume in the short term) or variable (scale per trade or per user). The major cost drivers tie directly back to the Financial Documentation: fixed personnel costs map to the CapEx team table during build and to the OpEx support staff table at run; variable cloud and API costs map to the OpEx scaling trajectory; and the contingency reserve lives in CapEx only. This block deliberately does not repeat dollar figures — those live in the Financial Doc — but every category here has a corresponding line item there.

Table 2.5. Cost Structure — Fixed vs Variable

Category	Type	Driver	Financial Doc Reference
Engineering team salaries (build)	Fixed	Headcount × duration	CapEx Table 3.1 — Personnel
Cloud infrastructure (baseline)	Fixed	Reserved instances, DB	OpEx Table 3.2 — Cloud hosting
Cloud infrastructure (scaling)	Variable	Trade volume, storage	OpEx Table 3.2 — Cloud hosting
KYC / AML API (Smile Identity)	Variable	Per verification	OpEx Table 3.2 — Third-party APIs
LBMA price feed	Fixed	Annual subscription	OpEx Table 3.2 — Third-party APIs
GoldBod / PMMC / BoG integration maintenance	Fixed	Annual compliance retainer	OpEx Table 3.2 — Compliance maintenance
Escrow bank integration fee	Variable	Per trade	OpEx Table 3.2 — Banking integration
Support staff (Tier 1 + success)	Fixed	Headcount	OpEx Table 3.2 — Support & success staff
Compliance audit (SOC 2, ISO 27001)	Fixed	Annual attestation	OpEx Table 3.2 — Compliance maintenance
Legal counsel (Ghana-barred)	Fixed	Retainer	OpEx Table 3.2 — Ongoing licenses & legal

Marketing (Chamber-co-funded)	Fixed	Quarterly campaign	OpEx Table 3.2 — Marketing
Contingency reserve	Fixed	15% of CapEx	CapEx Table 3.1 — Contingency

Table 2.5. Every cost category has a Financial Doc reference. The BMC's Cost Structure block and the Financial Documentation are intentionally redundant views of the same numbers. GoldBod/PMMC/BoG integration maintenance is now an explicit fixed-cost line.